Table Structures

Table 1: admins

Stores administrator account information for system access.

Field	Type	Null	Key	Default	Description
id	INT	NO	PRI	AUTO_INCREMENT	Primary key
username	VARCHAR(255)	NO	UNI	-	Admin username for login
password	VARCHAR(255)	NO	-	-	Hashed password
email	VARCHAR(255)	YES	-	NULL	Admin email address
name	VARCHAR(255)	YES	-	NULL	Admin full name

Indexes:

- PRIMARY KEY: `id`
- UNIQUE KEY: `username`

Relationships: None (standalone table)

Table 2: blood_groups

Stores blood group types and their details.

Field	Type	Null	Key	Default	Description
id	INT	NO	PRI	AUTO_INCREMENT	Primary key
name	VARCHAR(20)	NO	UNI	-	Full name (e.g., "A Positive")
code	VARCHAR(15)	NO	UNI	-	Blood type code (e.g., "A+")
description	TEXT	YES	-	NULL	Additional information

created_at	TIMESTAMP	YES	-	NULL	Record creation timestamp
updated_at	TIMESTAMP	YES	-	NULL	Record update timestamp

Indexes:

- PRIMARY KEY: `id`
- UNIQUE KEY: `name`
- UNIQUE KEY: `code`

Relationships:

- One-to-Many with `donors` (fk_blood_group_id)
- One-to-Many with `blood_inventory` (fk_blood_group_id)
- One-to-Many with `blood_distributions` (fk_blood_group_id)

Table 3: doctors

Stores doctor information for appointment management.

Field	Type	Null	Key	Default	Description
id	INT	NO	PRI	AUTO_INCREMENT	Primary key
name	VARCHAR(100)	NO	-	-	Doctor's full name
address	VARCHAR(255)	YES	-	NULL	Doctor's address
mobile	VARCHAR(20)	YES	-	NULL	Contact number
created_at	TIMESTAMP	YES	-	NULL	Record creation timestamp
updated_at	TIMESTAMP	YES	-	NULL	Record update timestamp

Indexes:

- PRIMARY KEY: `id`

Relationships:

- One-to-Many with `appointments` (fk_doctor_id)

Table 4: donors

Stores donor personal information, health data, and authentication details.

Field	Type	Null	Key	Default	Description
id	INT	NO	PRI	AUTO_INCREMENT	Primary key
fk_blood_group_id	INT	YES	FK	NULL	Reference to blood group
name	VARCHAR(100)	NO	-	-	Donor's full name
country	VARCHAR(50)	YES	-	NULL	Country of residence
address_line_1	VARCHAR(255)	YES	-	NULL	Primary address line
address_line_2	VARCHAR(255)	YES	-	NULL	Secondary address line
mobile	VARCHAR(20)	YES	-	NULL	Contact number
email	VARCHAR(100)	YES	-	NULL	Email address (unique)
password	VARCHAR(255)	NO	-	-	Hashed password
birthdate	DATE	YES	-	NULL	Date of birth
last_donation_date	DATE	YES	-	NULL	Date of last donation
hemoglobin_level	DECIMAL(5,2)	YES	-	NULL	Hemoglobin level (g/dL)
systolic	INT	YES	-	NULL	Systolic blood pressure

diastolic	INT	YES	-	NULL	Diastolic blood pressure
created_at	TIMESTAMP	YES	-	NULL	Record creation timestamp
updated_at	TIMESTAMP	YES	-	NULL	Record update timestamp

Indexes:

- PRIMARY KEY: `id`
- FOREIGN KEY: `fk_blood_group_id` references `blood_groups(id)`
- INDEX: `email` (for search performance)
- INDEX: `mobile` (for search performance)

Constraints:

- `birthdate`: Must be at least 18 years ago (application-level validation)
- `hemoglobin_level`: Range 0-20 (application-level)
- `systolic`: Range 70-200 (application-level)
- `diastolic`: Range 40-130 (application-level)

Relationships:

- Belongs to `blood_groups` (fk_blood_group_id)
- One-to-Many with `appointments` (fk_donor_id)
- One-to-Many with `blood_inventory` (fk_donor_id)

Table 5: patients

Stores patient information for blood distribution requests.

Field	Type	Null	Key	Default	Description
id	INT	NO	PRI	-	Primary key (manual input)
name	VARCHAR(100)	NO	-	-	Patient's full name
email	VARCHAR(100)	NO	UNI	-	Email address

medical_history	TEXT	YES	-	NULL	Medical history notes
address	VARCHAR(255)	YES	-	NULL	Patient's address
last_blood_taking_date	DATE	YES	-	NULL	Last blood transfusion date
created_at	TIMESTAMP	YES	-	NULL	Record creation timestamp
updated_at	TIMESTAMP	YES	-	NULL	Record update timestamp

Indexes:

- PRIMARY KEY: `id`
- UNIQUE KEY: `email`

Relationships:

- One-to-Many with `blood_distributions` (fk_patient_id)

Table 6: appointments

Manages donor appointments with doctors.

Field	Type	Null	Key	Default	Description
id	INT	NO	PRI	AUTO_INCREMENT	Primary key
fk_donor_id	INT	YES	FK	NULL	Reference to donor
fk_doctor_id	INT	YES	FK	NULL	Reference to doctor
appointment_time	DATETIME	YES	-	NULL	Scheduled appointment time
status	VARCHAR(20)	YES	-	'Pending'	Status: Pending/Confirmed/Cancelled/Completed

created_at	TIMESTAMP	YES	-	NULL	Record creation timestamp
updated_at	TIMESTAMP	YES	-	NULL	Record update timestamp

Indexes:

- PRIMARY KEY: `id`
- FOREIGN KEY: `fk_donor_id` references `donors(id)`
- FOREIGN KEY: `fk_doctor_id` references `doctors(id)`
- INDEX: `status` (for filtering)
- INDEX: `appointment_time` (for date range queries)

Constraints:

- `status`: ENUM type values (Pending, Confirmed, Cancelled, Completed)

Relationships:

- Belongs to `donors` (fk_donor_id)
- Belongs to `doctors` (fk_doctor_id)

Table 7: blood_inventory

Tracks collected blood units and their availability.

Field	Type	Null	Key	Default	Description
id	INT	NO	PRI	AUTO_INCREMENT	Primary key
fk_blood_group_id	INT	YES	FK	NULL	Reference to blood group
fk_donor_id	INT	YES	FK	NULL	Reference to donor
quantity	INT	YES	-	NULL	Amount in milliliters (ML)

collection_date	DATE	YES	-	NULL	Date of blood collection
expiry_date	DATE	YES	-	NULL	Expiry date (typically 35-42 days)
created_at	TIMESTAMP	YES	-	NULL	Record creation timestamp
updated_at	TIMESTAMP	YES	-	NULL	Record update timestamp

Indexes:

- PRIMARY KEY: `id`
- FOREIGN KEY: `fk_blood_group_id` references `blood_groups(id)`
- FOREIGN KEY: `fk_donor_id` references `donors(id)`
- INDEX: `expiry_date` (for expiry monitoring)
- INDEX: `quantity` (for availability checks)

Constraints:

- `expiry_date` > `collection_date` (application-level)
- `quantity` >= 0 (application-level)

Relationships:

- Belongs to `blood_groups` (fk_blood_group_id)
- Belongs to `donors` (fk_donor_id)

Table 8: blood_distributions

Records blood distribution requests and approvals.

Field	Type	Null	Key	Default	Description
id	INT	NO	PRI	AUTO_INCREMENT	Primary key
fk_patient_id	INT	YES	FK	NULL	Reference to patient

fk_blood_group_id	INT	YES	FK	NULL	Reference to blood group
request_unit	INT	YES	-	NULL	Requested amount (ML)
approved_unit	INT	YES	-	NULL	Approved amount (ML, NULL=pending, 0=rejected)
created_at	TIMESTAMP	YES	-	NULL	Record creation timestamp
updated_at	TIMESTAMP	YES	-	NULL	Record update timestamp

Indexes:

- PRIMARY KEY: `id`
- FOREIGN KEY: `fk_patient_id` references `patients(id)`
- FOREIGN KEY: `fk_blood_group_id` references `blood_groups(id)`
- INDEX: `approved_unit` (for status filtering)

Constraints:

- `approved_unit` <= `request_unit` (application-level)
- `request_unit` >= 1 (application-level)

Relationships:

- Belongs to `patients` (fk_patient_id)
- Belongs to `blood_groups` (fk_blood_group_id)

---

3. Table Relationships Summary

Table	Foreign Key	References Table	Relationship Type
donors	fk_blood_group_id	blood_groups.id	Many-to-One
appointments	fk_donor_id	donors.id	Many-to-One
appointments	fk_doctor_id	doctors.id	Many-to-One

blood_inventory	fk_blood_group_id	blood_groups.id	Many-to-One
blood_inventory	fk_donor_id	donors.id	Many-to-One
blood_distributions	fk_patient_id	patients.id	Many-to-One
blood_distributions	fk_blood_group_id	blood_groups.id	Many-to-One

---

4. Data Types Summary

Data Type	Used In	Purpose
INT	IDs, quantities, blood pressure	Numeric identifiers and counts
VARCHAR	Names, codes, addresses	Text strings with length limits
TEXT	Descriptions, medical history	Long text content
DATE	Birthdates, donation dates	Calendar dates without time
DATETIME	Appointment times	Date and time together
TIMESTAMP	created_at, updated_at	Automatic timestamps
DECIMAL	Hemoglobin level	Precise decimal values

---

6. Database Statistics

Table	Estimated Records	Growth Rate	Indexes
blood_groups	8 (fixed)	Static	2
doctors	10-50	Low	1
donors	1,000-10,000	Medium	2
patients	1,000-5,000	Medium	2
appointments	5,000-20,000	High	4
blood_inventory	2,000-10,000	Medium	3
blood_distributions	5,000-15,000	High	3

---

7. Notes for Database Implementation

1. **Foreign Key Constraints**: All foreign keys use `ON DELETE SET NULL` behavior (implicit in Laravel)
2. **Indexing Strategy**: Additional indexes recommended for frequently searched fields:
   - `donors.email` (unique)
   - `donors.mobile`
   - `appointments.appointment_time`
3. **Data Integrity**: Application-level validation ensures:
   - Age requirement (18+ years)
   - Donation interval (90 days)
   - Blood pressure and hemoglobin ranges
4. **Timestamps**: All tables use Laravel's automatic `created_at` and `updated_at` management
5. **Character Set**: UTF8MB4 recommended for full Unicode support